



General Instructions

The assignment concerns the test-driven development and integration testing of an autonomous parking assistance system (https://en.wikipedia.org/wiki/Automatic_parking). From the testing and verification perspective, the project comprises the application of the following techniques:

- **Test-Driven Development**,
- **Unit Testing** using `jUnit`,
- gathering **Coverage Metrics** using `EclEmma` (<http://www.eclemma.org/>) (or similar tools),
- **Integration Testing**, including developing **stubs** using `Mockito` (<http://site.mockito.org/>) (or similar tools), alternatively doing **manual mock-ups**.

The very core of this project is the application of Test-Driven Development (**TDD**), after some basic lectures on testing theory in the first week of the course, in the second week we will have a 2 x 3 hours module on TDD given by a specialist in industry. **Note!!!** that this module has no recordings of any kind, nor will it be recorded this year, so attending the lectures is the only way of familiarizing yourself with the material (the lecture slides and examples will be available after the lecture though).

This is a group project that is to be carried out in groups of **3-4**. You need to have formed your groups and enroll on Blackboard by **Monday September 7th End of Day**. The deliverables are to be submitted on Blackboard (submission entries to be found below), and towards the end of the testing module (starting **October 5th**) of the course each group is going to be examined orally with individual questions related to the project or the associated course contents. The exact schedule for this will come in due time. The deliverables typically comprise a **report and the implementation code**: the report should document the major steps in each phase and code snippets (few concise examples) of how they are implemented and possibly screenshots of the results. Extensive pieces of program code should not be included in the report. The most important factors in judging the reports are: their logical structure and sufficiently clear explanation of the steps (all figures and code snippets should be accompanied with clear descriptions).

In the schedule, there are three slots of **two hours of face-to-face supervision** offered for each group, check the schedule for the exact time. You are expected to come prepared with your questions and problems to these sessions or with ready solutions to show.

The deadlines are to be respected for each phase. You may get an **extension of one week for each phase**; for that you need to send an email before the deadline to Wojciech. Sending an email is sufficient for receiving the extension and you do not need to wait for a response to your email.

As a general principle, when you find an ambiguity in the requirements, make a reasonable assumption and document it clearly in your report. Another tip is - do not over-complicate things, especially in the first phase of the project when you can feel the urge to include things outside of the phase's specification just because you think that some things are missing (it is very likely they will come in the second phase).

The assignment is organized in the two following phases:

- **Phase 1: *TDD and Unit Testing***, Deadline: **September 18, 2026, EOD**
- **Phase 2: *Integration Testing***, Deadline: **October 2, 2026, EOD**